

Consequences of Arrow

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Group Activity Answers

At the end of last class, we had a group activity about three voting systems:

1. **Plurality:** Candidates get 1 point per first-place vote.
2. **Range voting:** Each voter gives 1-5 star ratings, candidates ranked by average.
3. **Approval voting:** Voters approve/disapprove each candidate, candidates ranked by approval count.

We wanted to go over which of Arrow's conditions each failed.

Arrow's Theorem: There is no social choice function that simultaneously satisfies:

- Unrestricted Domain (U)
- Weak Pareto Principle (P)
- Independence of Irrelevant Alternatives (I)
- Non-dictatorship (D)

Question: Which Arrow condition does plurality violate?

Answer: Both I and P

We already covered one way plurality violates I last time—the example where people changing their minds about B vs C affected the ranking of A vs B.

But plurality also violates the Pareto principle. Imagine 100 voters choosing among A, B, C, and D.

- Everyone thinks A is second-worst and B is worst.
- But their first choices are split among C and D.

The result is that A and B are tied at 0 votes each, so $A \sim B$ in the social ranking.

But Pareto says that if everyone prefers A to B, then $A \succ B$.

In practice the failure of I gets talked about more than the failure of P, because we don't normally care about how voting systems rank unselected candidates.

Question: Which Arrow condition does range voting violate?

Answer: First, it's not even a function, so it isn't even eligible to be an Arrow-style social choice function. It also manages to violate both I and P.

Range voting violates the basic setup assumption that the social ranking is a function of individual preferences.

Same preferences can produce different social rankings if strength of preference matters. We'll come back to that below.

Since the social ranking depends on numerical scores, not just preferences, range voting violates I.

Here is a really simple example. Imagine that there are two voters, Mitch and Brian, and two candidates, A and B.

Table 1: Monday (B wins)

	Mitch	Brian	Average
A	3	1	2
B	1	4	2.5

Table 2: Tuesday (A wins)

	Mitch	Brian	Average
A	5	1	3
B	1	4	2.5

No one changed their comparative preference between Monday and Tuesday, but the social ordering changed. That's what I rules out.

As some of you pointed out in the groups, range voting can also violate Pareto.

- Everyone gives A and B five stars (much better than the rest)
- Everyone also prefers A to B

The result is that the social ranking has $A \sim B$ (tied at 5 stars).

But Pareto says that if everyone prefers A to B, then $A \succ B$.

Question: Which Arrow condition does approval voting violate?

Answer: Both I and P, for essentially the same reasons as range voting.

Approval voting has the same problems as range voting:

1. **Not functional:** Same preferences can yield different results depending on who voters choose to “approve”
2. **Violates I:** Changing approval of C can change the A vs B ranking
3. **Violates P:** Everyone could prefer A to B but approve both equally, giving $A \approx B$ instead of $A \succ B$

All three systems violate multiple Arrow conditions:

- **Plurality:** Violates I and P
- **Range voting:** Not functional, violates I and P
- **Approval voting:** Not functional, violates I and P

This illustrates Arrow's point: every real voting system must give up something.

Consequences

You'll often read people saying that Arrow's Theorem is a particular problem for **democratic** systems.

It seems to show that there is no system of aggregating votes that meets some really plausible conditions.

So is that bad news for voting systems?

- Yes, but...

Let's say that instead of democracy, your preferred social model is benevolent dictatorship, where an all-powerful ruler acts in the best interests of the people collectively.

- Aside one: If this were a history of social thought class, we'd walk through how popular a model this was among reformers through the 19th century.
- Aside two: For things where the government has to act although approximately zero votes will turn on how they act, this is our actual system. Think about things like the city council deciding the rules for a particular intersection.

If our dictator genuinely cares about the people, then what they do should be sensitive to their interests collectively. And that presumably means:

1. They have to act no matter what those interests are (U)
2. They should do A rather than B if it's in everyone's interests (P)
3. Whether they do A or B should only turn on which of those two is better for the people (I)
4. They shouldn't act just in the interests of one person (D)

Note that the condition D does not just say there shouldn't be one person making all the decisions.

It says that the decisions shouldn't be made solely based on what one person wants, or in the interests of one person.

So really all the conditions might need to be met, even for a benevolent dictator.

Well maybe you don't like dictators; I don't particularly either.

Perhaps you think a market system of free trade between free people will produce a better outcome for the people collectively.

That's an interesting claim, and actually one that Arrow himself did some important work on formalizing.

But we still need to define what it is for something to be a better outcome.

And we'll now have the same problem the benevolent dictator faced.

- Our theory of better outcomes should work no matter what's good or bad for individuals (U)
- What's better for everyone should be better overall (P)
- Whether A or B is better overall should just turn on which of them is better for individuals (I)
- Whether A or B is better overall should not just turn on what's better for one person (D)

Arrow's Theorem does raise a challenge to democratic systems.

But it raises more or less the same challenge for non-democratic systems as well.

What it really challenges is the idea of a public good, and that's something that democrats, benevolent dictators, and free marketeers all care about.

Two Partial Ways Out

I won't spend much time on this, but it turns out that there are some pretty interesting results on how far you can go if you restrict U.

The most significant result concerns what are called **single-peaked preferences**.

Imagine that there is some linear scale you can place the options on, and for every voter, the options satisfy the following condition:

- The voter has a preferred spot on the scale.
- If A is between B and that preferred spot, then the voter prefers A to B.
- E.g., all the candidates fall on a left-right spectrum, and so do all the voters, and a voter will never prefer to move away from their preferred spot on the spectrum.

Given this constraint, majority rule works. You never get Condorcet-style cycles.

- Is this realistic?
- It sometimes is! If you think in advance that your voter pool meets this condition, simple majority rule is a perfectly good rule.
- In a different class I'd work through a more precise statement of what single-peaked means, because it's actually a little more generous a condition than it looks, but we'll leave it here unless people have particular questions.

Amartya Sen, who we'll talk more about in future lectures, argued for a while that we should give up the constraint that the output should be an ordering.

In particular, he thought that we should give up the constraint that the output should be complete.

So let's say that we want an output to be a pair $>_{\forall}$ and $=_{\forall}$, and our rule is unanimity.

If everyone prefers A to B, then $A >_{\forall} B$, and if everyone ranks them equally, then $A =_{\forall} B$.

We still insist that at most one of $A >_{\forall} B$, $B >_{\forall} A$, and $A =_{\forall} B$ holds, but we don't insist that at least one holds.

This rule satisfies all our constraints, except for completeness of the social ordering:

- It gives a result whatever the input, satisfying U.
- If everyone prefers A to B, so does the social ordering, satisfying P.
- The social ordering of A and B only turns on what people think about those options, satisfying I.
- And anyone might not get their way on A and B, satisfying D.

The problem is that this is an incredibly weak rule.

- It just says, choose Pareto-dominant options.
- It doesn't say what to do when none of them exist.
- Moreover, dropping completeness essentially only gets you this extra rule.
- Allan Gibbard (a longtime member of the philosophy department at UM) proved that the only rule satisfying all four constraints but not completeness is unanimity among 'oligarchs', where membership of the oligarchy could be anything.

Independence of Irrelevant Alternatives

Here's a research topic that at one time I thought would be interesting:

- Look through actual existing voting systems.
- Since they exist, they must be possible.
- And so they must violate one of the Arrow constraints.
- Question: For each system, which one do they violate?

- They violate Independence of Irrelevant Alternatives (I).
- They **all** violate I.
- It wasn't an interesting research topic because that's always the answer.

When you see weird failures of plurality, like when two similar candidates lose because of vote splitting, it's tempting to say that the problem is that plurality voting violates I.

- That can't be right on its own; all systems violate I.
- What could be a good complaint is that plurality leads to *frequent* violations of I in *realistic* situations, but the existence of violations can't be the problem.

Here's one difference between IIA and real-life voting systems:

- Voting systems tend to care a lot about which option a voter ranks **first**. (Plurality says this is all that matters, which is extreme, but all real-life systems have this feature.)
- The spirit behind IIA is that this shouldn't matter.
- If I prefer A to B, that should count the same whether or not A is my **first** choice.

So one question to think about is basically: does being a voter's first choice matter, or is this just another spot on the preference ranking?

I think there's a reasonable case that being first should matter.

- Think about football fandom (to use a timely but non-political example).
- It matters a lot more to most people whether their most preferred team (i.e., the team they're a fan of) wins, than whether it's their 12th or 13th favorite team that wins.
- That's the intuition behind voting systems, and I think it's an OK one.

Numerical Measures

Another way aggregation systems get around Arrow is by using numerical scales:

- Amazon reviews average out numbers from 1-5.
- Traffic planners try to minimize everyone's time spent at an intersection (conditional on the intersection being safe).
- How does this interact with the Arrow principles?

If anything, things are now **worse**. This isn't even a system for converting individual preferences to social preferences.

- Arrow says that the aggregation system should take the preferences of the voters as an input, and return a social ordering as the output.
- But numerical measures of how good an option is for a citizen/voter can't be deduced from their preferences.

In the two tables below, Alex and Charlie are 'voters', and the numbers show how they rate (on a 1-5 scale) blickets and widgets. The left table is what they think on Monday, the right table is Tuesday.

Table 3: Monday's ratings of blickets vs. widgets

Voter	Blicket	Widget
Alex	5	1
Charlie	2	3
Average	3.5	2

Table 4: Tuesday's ratings of blickets vs. widgets

Voter	Blicket	Widget
Alex	3	2
Charlie	1	5
Average	2	3.5

Table 5: Monday's ratings

Voter	Blicket	Widget
Alex	5	1
Charlie	2	3
Average	3.5	2

Table 6: Tuesday's ratings

Voter	Blicket	Widget
Alex	3	2
Charlie	1	5
Average	2	3.5

- On Monday, blickets have a higher average score than widgets.
- On Tuesday, it's the other way around.
- But no one's preferences changed—both Alex and Charlie prefer blickets to widgets on both days.

Technically, this is a violation of I.

But more importantly, it's a violation of functionality.

The social ordering is not a function, in the mathematical sense, of the preferences of the voters.

Any numerical system will be like this.

So why isn't this the solution? Why don't we just use numerical measures of how good an option is for each person?

- One reason is one we discussed in voting systems: it really doesn't work in voting because the incentives to vote 'strategically' are too high and too easy to spot.
- But another reason comes from philosophy.

If the benevolent dictator wants to use numbers not rankings, they have to generate what are called “Interpersonal Utility Comparisons”.

The standard view in mid-century philosophy/economics/social science was that these were impossible.

The canonical argument for this is Lionel Robbins’s argument in his 1932 book *An Essay on the Nature and Significance of Economic Science*, which turned on the difficulty of measuring comparisons between people.

For the benevolent dictator, or the free market exponent, this turns out to be the big question.

If you can say not just that A is better than B for citizen C, but you can measure how much better it is, then you should use that quantity in making social decisions.

If not, you're back in Arrow's framework, and you have to think about which condition to give up.

If we have time, we'll discuss this more, but here's one important takeaway from Arrow.

- There is no perfect voting system.
- So, don't assume that the best voting system for one task is the best one for another.
- In particular, the best systems for small groups and large groups will probably be different.

We'll look at a surprising argument that sometimes the right condition to give up is the Pareto optimality condition.

Some plausible-looking systems of social choice end up violating it, and perhaps that's a good thing.